Tertiary Infotech Pte Ltd
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LESSON PLAN

For

AI Security for Autonomous AI Agents

TGS Ref No: TGS-2025060473

Conducted by

Tertiary Infotech Pte Ltd
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Version 4.13

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	16 June 2025	Initial release.	Dr Alfred Ang
2.0	17 August 2026	Retitled and rebuilt for autonomous-agent security; five activities added. TSC K/A text unchanged.	Dr Alfred Ang
2.1	18 August 2026	Added prompt-injection, PDPA, guardrail, approval, skill/plugin and visual coverage. TSC K/A text unchanged.	Dr Alfred Ang
3.0	20 August 2026	Evidence-grounded 207-slide rebuild with product boundaries, cases, controls, activities and source register. TSC K/A text unchanged.	Dr Alfred Ang
4.0	25 August 2026	Rebuilt as a 1-day course with three LO-aligned topics, eight no-code activities and a reflection-based practical assessment. TSC K/A text unchanged.	Dr Alfred Ang
4.1	25 August 2026	Refreshed trainer visuals and embedded the YouTube Online Video; scope, timing and TSC K/A text unchanged.	Dr Alfred Ang
4.2	25 August 2026	Added coding-harness and latest-AI-agents reference slides to the deck (Claude Code, Codex, DeepSeek Harness; OpenClaw, Hermes, Prime Agent, OpenWorker, QM). Schedule and TSC K/A text unchanged.	Dr Alfred Ang
4.3	25 August 2026	Beefed up Topic 3 with Singapore AI case studies (adoption vs cybersecurity, gen-AI brand damage, agentic testing, healthcare AgentSea), a brand-damage discussion/debrief, a 'Key Challenges of AI to Business' capstone	Dr Alfred Ang

		and a visual multi-agent map. Schedule and TSC K/A text unchanged.	
4.4	25 August 2026	Activity 5 rebuilt around the AI Data Policy Generator website (learner-supplied training API key, 7-part IMDA/PDPA-aligned policy framework). Slides, LG and activity pack aligned; added a worked-example slide with concrete Scope, Roles, Rules and Review examples. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.5	25 August 2026	Added Topic 1 slide 'Probabilistic Output — Why Traditional Security Struggles' contrasting deterministic software with probabilistic LLM output and the challenge to rule-based defences. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.6	25 August 2026	Added Topic 1 slide 'When the Loop Goes Wrong — Misreading the Goal' after the concrete agentic-loop example: an open-ended 'remove the malware at all costs' goal escalates to wiping every file, motivating the Topic 3 guardrails. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.7	25 August 2026	Added Topic 1 slide 'An Agent on WhatsApp Is Exposed to the Internet' after the TIA Support slide: a public messaging agent accepts messages from anyone, so prompt injection and data poisoning can extract	Dr Alfred Ang

		confidential data or harm the server. Schedule timing and TSC K/A text unchanged.	
4.8	26 August 2026	Added verified case study 'Case — A Malicious Skill on ClawHub' (Snyk, Feb 2026) at the end of the Topic 1 Skills and Tools section: a fake Google skill whose SKILL.md instructions tricked users into installing malware. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.9	26 August 2026	Activity 3 reworked: the agent now creates a 10-slide 'AI Security for AI Agents' deck (saved to the Downloads folder), then redesigns it from an uploaded image template — replacing the Envato-template animated-deck flow. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.10	26 August 2026	Activity 2 reworked: learners now upload MOCK-MARKETING-DATA.xlsx to the Hermes + MiniMax agent and prompt it to analyse the data, generate charts and produce a .docx report with insights, analysis and recommendations. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.11	26 August 2026	Added four Topic 1 slides on applications of AI agents in cyber security: threat detection (real-time detection, automated vulnerability analysis, malware classification), network and social-engineering security (intrusion detection, attack	Dr Alfred Ang

		classification, phishing defence, log anomaly detection), incident response (patch generation, playbooks, root-cause analysis) and Security Operations Centres (threat hunting, predictive analytics, adaptive decision-making). Learner Guide notes renumbered. TSC K/A text unchanged.	
4.12	26 August 2026	Added Topic 1 slide 'Agent-to-Agent Messaging - A Security Blind Spot' after the multi-agent worked example: bad or malicious output cascades agent-to-agent, and a growing topology multiplies message paths - reducing visibility and making data-flow tracing and attribution harder. Beefed up Topic 3 security content with ten slides woven into the existing sections: multi-layer bias mitigation; the four structural security gaps of agents (multi-step inputs, tool chaining, opaque execution, untrusted entities); direct vs indirect prompt injection; jailbreaking and jailbreak persistence; a new 'Advanced Threats — A Look Ahead' section (data poisoning, backdoor attacks, AI supply-chain risk, agent-to-agent attacks, misalignment / Goodhart's Law); and human-in-the-loop at machine speed. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.13	26 August 2026	Practical instrument	Dr Alfred Ang

		renamed from Practical Performance to Case Study across the assessment set, slides, Lesson Plan and Learner Guide. Each Case Study task trimmed to TWO questions: Task 1 keeps the generative-vs-agentic-vs-agent and useful-vs-not-trusted questions; Task 2 keeps the weak-vs-strong prompt and prompt-principles questions; Task 3 keeps the leaky-vs-guarded chatbot and accountability questions. Marks rebalanced (20/24/26, total 70). Schedule timing and TSC K/A text unchanged.	

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1. Course Overview

AI Security for Autonomous AI Agents is a 1 Day · 8 Hours WSQ course that introduces generative AI, agentic AI and autonomous AI agents, and the security, governance and ethical risks they bring to customer-service and hospitality settings. Learners work entirely on ready-made websites, chatbots and AI agents — no coding is required. The day is organised as three topics mapped one-to-one onto the three learning outcomes.

Topic 1 (LO1) explains how generative AI works and how it becomes agentic — the agentic loop, harness engineering, AI agents such as OpenClaw and Hermes, skills, tools and multi-agent systems. Topic 2 (LO2) covers prompt engineering and post-training, using the Hermes agent on a MiniMax model to analyse data and build presentations. Topic 3 (LO3) covers AI data governance, job impact and redesign, and the cybersecurity risks of autonomous agents, including a safe roll-out framework. Real, dated examples and clearly labelled simulations are grounded in current governance references such as the IMDA Model AI Governance Framework, Singapore's PDPA and PDPC guidance, and the OWASP Top 10 for LLM and Agentic Applications.

Skills Framework Alignment

Element	Detail
TSC Title	Generative AI Principles and Applications
TSC Code	ICT-INT-0052-1.1
Proficiency Level	Level 1
Course Code	TGS-2025060473
Duration	1 Day · 8 Hours
Delivery	Instructor-led classroom with facilitated case-study activities

Note: the accredited knowledge and ability statements below are reproduced verbatim. They are taught and assessed as accredited; AI security is the delivery context through which each statement is evidenced.

Learning Outcomes

LO	Learning Outcome
LO1	Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
LO2	Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
LO3	Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

Knowledge Statements (assessed by the Written Assessment)

Code	Knowledge statement
K1	Importance of data quality, preprocessing, model pipeline and model training (e.g., impact of data bias from training data)
K2	Underlying principles, core concepts and theories governing generative AI
K3	Difference between generative and discriminative models
K4	Impact of prompt engineering on the model outputs of generative AI
K5	Generative AI model workings, including training data, algorithms, and outputs

Ability Statements (assessed by the Case Study)

Code	Ability statement
A1	Analyse limitations and potential biases in AI-generated content
A2	Identify the ethical implications and societal impact of AI-generated content
A3	Apply understanding of generative AI principles to use cases
A4	Demonstrate the use of generation AI in diverse applications (e.g., summarisation, inference, reasoning, transformation of content, augmentation of content)
A5	Analyse generative AI models' performance metrics and evaluate the influence of prompt variations

2. Course Schedule

The training day runs 9:00am – 6:00pm and delivers 8 instructional hours. A 1-hour lunch break is excluded from instructional time; short tea breaks are counted within the day. The 1-hour assessment runs 6:00pm – 7:00pm.

One-Day Schedule (Topics 1–3)

Slides 1–165 of the trainer deck.

Time	Topic / Activity	Duration
9:00 – 9:20	Welcome, digital attendance, introductions, learning outcomes and ground rules	20 min
9:20 – 10:05	Topic 1: Brief history of AI 2023-2026; how generative AI works — autoregressive LLM, training and inference (K2, K5)	45 min
10:05 – 10:45	Topic 1: Real generative-AI use cases (AI fashion models, AI video, Klarna); context engineering (K3, A3, A4)	40 min
10:45 – 11:00	Tea break	15 min
11:00 – 11:45	Topic 1: Agentic loop and harness engineering; AI agents, OpenClaw and Hermes, skills, tools and multi-agent systems (K2, K3)	45 min
11:45 – 12:30	Activity 1: Talk to the TIA Support AI agent on WhatsApp and post reflections to the Pinboard; debrief (A2, A4)	45 min
12:30 – 1:30	Lunch break	60 min
1:30 – 1:45	Topic 1 summary: generative AI vs agentic AI vs AI agents	15 min
1:45 – 2:20	Topic 2: Install the Hermes agent on MiniMax M2.7; principles of prompt engineering, good vs bad prompts (K4, A5)	35 min
2:20 – 3:10	Activity 2 & 3: Upload mock Excel data and prompt the agent into charts and a .docx insights report, then build a 10-slide PPT and redesign it from an uploaded image template (K4, A5)	50 min
3:10 – 3:25	Tea break	15 min
3:25 – 3:50	Activity 4: Install tools and skills, re-run the Excel and PPT tasks; Topic 2 reflection and debrief (A1, A5)	25 min
3:50 – 4:35	Topic 3: AI data governance and accountability (Moffatt v Air Canada); Activity 5 — generate an AI data governance policy with the policy-generator website (A1, A2)	45 min

4:35 – 5:15	Topic 3: Job impact and redesign; Activity 6 — coaching role-play simulator (A2)	40 min
5:15 – 6:00	Topic 3: AI-agent cybersecurity risks, advanced threats (prompt injection, jailbreaking, data poisoning, supply-chain and agent-to-agent attacks) and safe rollout; Activities 7 & 8 — leaky vs guarded chatbots and security reflection (A1, A2)	45 min
6:00 – 6:35	Briefing for assessment; Written Assessment (SAQ) — 5 questions, one per K statement (K1-K5)	35 min
6:35 – 7:00	Case Study — 3 reflection tasks (two questions each) on the activities completed; TRAQOM close	25 min

Instructional total: 480 minutes (8 hours), excluding the 1-hour lunch break and the 1-hour assessment.

3. Activities

Every activity uses a ready-made website, chatbot or AI agent — learners do not write any code. Each has its own folder under activities/ containing learner instructions, prompt cards, mock data and printable PDFs. All data is fictional. Full step-by-step facilitation detail is in the Learner Guide.

#	Activity	Topic	Duration	LO / A
1	Talk to an AI Agent (TIA Support WhatsApp) and reflect on the Pinboard	1	45 min	LO1 · A2, A4
2	Analyse mock Excel marketing data into a charted .docx report with the Hermes + MiniMax agent	2	25 min	LO2 · A5
3	Create a 10-slide PPT with the agent, then redesign it from an uploaded image template	2	25 min	LO2 · A5
4	Install tools and skills, then re-run the Excel and PPT tasks	2	25 min	LO2 · A1, A5
5	Generate an AI Data Governance Policy with the policy-generator website	3	30 min	LO3 · A1, A2
6	Coach a worried team member (AI role-play simulator)	3	30 min	LO3 · A2
7	Leaky vs guarded chatbot — see and stop a PII leak	3	30 min	LO3 · A1, A2
8	Reflect on agent security and decide go / no-go	3	15 min	LO3 · A1, A2

Times overlap with the topic teaching blocks in the schedule above; activity folders and the reflection Pinboard (alfredang.github.io/pinboard) are used throughout the day.

4. Tools and Resources

Resource	Purpose
Trainer slide deck (PPTX/PDF)	Facilitation, diagrams and concept walkthroughs
Learner Guide	Detailed notes and step-by-step activity walkthroughs
Activity packs (activities/)	Ready-made websites/chatbots, prompt cards and mock data
TIA Support on WhatsApp (+65 8866 6375)	Live OpenClaw-powered AI agent for Activity 1
Reflection Pinboard	alfredang.github.io/pinboard — group reflections by risk theme
Hermes Agent + MiniMax	hermes-agent.nousresearch.com and minimax.io for Topic 2
LMS/TMS portal	lms-tms.tertiaryinfotech.com — course material and submission
IMDA / PDPC / OWASP references	Governance and security reference material

5. Assessment

Assessment is conducted at the end of the day (6:00pm – 7:00pm). The briefing for assessment is delivered before the assessment begins.

Instrument	Covers	Detail
Written Assessment (SAQ)	K1 – K5	Written Assessment (SAQ) — 5 short-answer questions, one per K statement (K1–K5)
Case Study	LO1 – LO3 (A1 – A5)	Case Study — 3 reflective tasks (two questions each) in which learners document their own observations from the in-class activities, mapped to LO1–LO3 and the A statements
Format	—	Open Book
Grading	—	Competent / Not Yet Competent
Re-assessment	—	Available for learners assessed Not Yet Competent

The Case Study asks learners to document their own observations and reflections from the activities they completed during the day, so it is grounded in what each learner actually did in class and aligned to the slides and labs.

Funding eligibility requires a minimum 75% attendance recorded through SSG digital attendance, an assessment outcome of Competent, and completion of the TRAQOM survey.